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Data–driven product form design integrating affective and functional requirements via TGRA–QFD and Stable Diffusion

Jiawei CHEN*, Zhiyong ZHOU*, Jianming QI* and Hanfei WANG*
*School of Design and Art, Shanghai Dianji University
No. 690, Jiangchuan Road, Shanghai 200240, China
E-mail: zhouzhiyong@sdju.edu.cn

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Abstract
The integration of affective and functional requirements has become a critical issue in achieving user-centered and competitive product design. Traditional approaches tend to focus on either affective or functional aspects in isolation, resulting in an incomplete understanding of user requirements. To address this gap, this study proposes an AI-driven product form design framework that simultaneously extracts and integrates affective requirements (ARs) and functional requirements (FRs) from online reviews. The research involves four stages. First, user reviews of the target product are mined to extract ARs and FRs. Then, the weights of the ARs are calculated based on the Rank Correlation Analysis (RCA), and an innovative Time-weighted Grey Relational Analysis (TGRA) method is proposed to determine the weights of the FRs by considering the influence of temporal factors. Second, Quality Function Deployment (QFD) maps the ARs and FRs to concrete design features (DF). Third, these features are converted into structured prompts for the Stable Diffusion Model (SDM) to produce product concept sketches. Finally, a VR eye-tracking experiment provides both sensory and cognitive evaluations to identify the optimal concept. A case study on an aerial-photography drone (APD) selects the best form using an aesthetic measurement formula and validates it through VR eye-tracking. The results indicate a strong consistency between the experimental findings and the subjective questionnaire outcomes, confirming the effectiveness of the proposed design model. This model enhances design efficiency while ensuring the visual aesthetics of the product.

Keywords: Product form design, Text mining, Kansei engineering, AI image generation, Unmanned aerial vehicle

1. Introduction

In the context of increasingly competitive product design, enterprises face the challenge of addressing diverse and dynamic user requirements (Dong et al., 2023) that encompass both rational and emotional dimensions. These requirements involve functional and affective dimensions, which respectively refer to the product's utility and users' perceptual or emotional experiences (Fu et al., 2024; Yan and Li, 2022). While product design success relies on effectively meeting these multidimensional needs (Wang et al., 2024), previous research has tended to examine affective and functional requirements separately, leading to an incomplete understanding of user expectations and a disconnection between user perception and product function. Besides, these requirements are often hidden and vague. Accurately grasping and satisfying user requirements presents a great challenge (Xu et al., 2023). To address this issue, this study emphasizes integrating Affective Requirements (ARs) and Functional Requirements (FRs) within a unified design framework to balance emotional appeal and function-related perceptual considerations at the product form level.

Kansei Engineering (KE) is a methodology that translates user requirements into design features (DF). It is widely applied to explore users' emotional characteristics and requirements (Liu et al., 2023). It enables designers to quantify users' emotional responses and transform abstract requirements into concrete DFs. Unlike traditional qualitative emotional design methods, KE quantitatively links abstract user requirements to DFs. This transformation relies on quantitative analysis of user requirements for the target product (Kang et al., 2025b). Traditionally, consumers' emotional

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responses are obtained through questionnaires and interviews (Du et al., 2024). However, designers' subjective experience alone cannot accurately capture users' psychological states, emotional needs, or aesthetic preferences (Yang et al., 2023). Consequently, recent studies employ data-driven approaches to acquire users' ARs and translate them into DFs, such as shape, material, and colour, to guide design decisions (Park and Hyun, 2022).

However, previous KE studies exhibit limitations in addressing complex perceptual rankings and constructing reliable mapping models (Kang et al., 2025a). Common computational approaches for weighting users' perceptual requirements include the Analytic Hierarchy Process (AHP), the Kano model, and the entropy weighting method. Each method, however, has inherent drawbacks when used independently: AHP adapts poorly to complex demand structures, the Kano model is overly sensitive to subjective judgments, and the entropy method fails to capture perceptual attributes adequately. Functionality represents the product's intended purpose (Z. Wang et al., 2025) and serves as a key determinant of customer purchase intention (Lee, 2024). However, current research in product appearance research has focused on extracting and analyzing ARs, while neglecting to include product functionality in design considerations. To address this gap, this paper proposes an online review‐based analysis method that calculates the weights and rankings of users' ARs and FRs and maps them to DFs. This method effectively combines online reviews and structured analysis, overcoming the problems of subjective-objective bias and insufficient consideration of product features in traditional methods, thus improving the accuracy and consistency of decision-making.

Although KE can map user requirements to DFs, concept generation remains highly dependent on designers, whose creativity is constrained by expertise and subjectivity (Yang et al., 2023). As a result, even experienced designers may fail to fully satisfy affective requirements, leading to repeated revisions and extended development cycles. With the rapid advancement of Artificial Intelligence (AI) technologies, this limitation has been substantially alleviated. An increasing number of studies have successfully integrated AI techniques into product design, enabling designers to generate renderings directly from specified DFs (Du et al., 2024; Yang et al., 2025). Among these techniques, Diffusion Models (DM) have gained significant attention for their ability to synthesize high-quality and diverse images across various design domains (Yang et al., 2023). Building on this, the SDM (Rombach et al., 2022) introduces a latent vector space that improves generation efficiency and image fidelity. By allowing designers to input textual or visual design requirements into pretrained models, SDM effectively lowers creative barriers, accelerates concept exploration, and improves the overall efficiency of the design process. Accordingly, this study introduces the SDM into KE. This image generation algorithm is an effective improvement on the efficiency enhancement of the traditional conventional design process. It demonstrates strong performance in the field of sketching as well as conceptual rendering generation. To further enhance SDM's creative capability, this study simultaneously maps ARs and FRs into DFs and transforms them into cues according to SDM prompt specifications. The resulting product concept sketches are subsequently designed and generated.

In summary, this study proposes a product form design framework that integrates user review analysis with artificial intelligence image generation. First, user comments are collected from online shopping platforms using a web crawler, and the text data are processed to extract affective and functional vocabulary. These vocabularies are subsequently transformed into users' ARs and FRs through KE. Second, QFD is used to construct a mapping model linking users' ARs and FRs and product DFs. Third, the DFs produced by the mapping model are transformed into prompts and entered into SDM to conceptualize sketch alternatives, which are then filtered and refined to produce final product renderings. Finally, the generated designs are evaluated using subjective scoring and objective physiological experiments to verify that the resulting target product form meets user expectations.

The remainder of this paper is organized as follows: Section 2 introduces the relevant literature. Section 3 proposes the research process framework. Section 4 presents a specific case of the APD form design, accompanied by experimental validation through eye-tracking methodology. The discussion and the future research directions are proposed in Section 5, and Section 6 summarizes and presents the contributions of this study.

2. Literature review

2.1 Integration of ARs and FRs in product design

Recent product design research increasingly emphasizes integrating ARs and FRs, as these dimensions jointly determine user satisfaction and product competitiveness. ARs reflect users' emotional and perceptual responses, whereas FRs concern users' perceptions of usability, functional plausibility, and practical associations inferred from product form.

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KE has been widely adopted to systematically capture users' emotional perceptions and translate them into DFs. It bridges affective cognition and concrete design parameters, providing a basis for integrating affective and functional perspectives.

Traditional KE-based studies mainly explored users' ARs through questionnaires or semi-structured interviews (Zhu et al., 2024). However, such analyses are limited by sample representativeness and expert subjectivity, reducing the depth of requirement extraction (Hsiao et al., 2017). To address this limitation, many scholars have applied KE to extract users' ARs from online reviews (Zheng et al., 2024). For instance, Woo et al .(2022) used Word2Vec, factor analysis, and hierarchical cluster analysis to extract perceptual words from reviews and guide product design . Chiu and Lin (2018) employed text mining and sentiment analysis of Amazon reviews to identify customer preferences and convert them into design parameters. These studies confirm that KE effectively captures users' ARs from online reviews and maps them to DFs.

Recent studies highlight the need to simultaneously satisfy ARs and FRs (Bing et al., 2025). Some studies incorporate functional attributes into AR analysis to guide or evaluate product design. Wang et al. (2024) proposed an affective PFBD method integrating functional analysis and emotional evaluation to generate and assess diverse conceptual design schemes. In a subsequent study, Wang et al. (2025) fine-tuned the Stable Diffusion Model (SDM) using LoRA based on affective and functional vocabularies and employed multi-stage fuzzy evaluation to select optimal designs satisfying both ARs and FRs . These studies underscore the necessity of integrating ARs and FRs to ensure product forms align with users' functional needs and aesthetic preferences.

2.2 Integration mechanisms for affective–functional weighting and QFD mapping

After identifying ARs and FRs, the next step is to quantitatively analyze their relative importance and establish a reliable mapping between user perceptions and DFs. Previous studies have typically used mathematical algorithms to investigate the relationships between key variables. Perceptual vocabularies are first extracted from user research or online reviews and analyzed in combination with word polarity and frequency to identify representative affective descriptors (Jin et al., 2022). Subsequently, analytical approaches such as the Analytic Hierarchy Process (AHP) (C. Wang et al., 2024) are employed to determine the relative weights of perceptual terms, enabling the construction of quantitative mapping models that link user perceptions to specific DFs. However, expert-based methods like AHP, despite their effectiveness in prioritizing user requirements, are inherently influenced by subjective judgments and consistency constraints. To overcome this limitation, Ren et al. (2023) employed RCA to convert adverbial expressions into degree features to quantify users' affective tendencies in online reviews. This method reduces expert subjectivity and provides a data-driven approach for objectively deriving AR weights and rankings.

Since FRs cannot be represented through adverbial expressions in user comments as ARs can, alternative methods are needed to calculate FR weights. In the field of requirements analysis, GRA is widely used due to its advantages, such as small sample size requirements, relatively low computational cost, and ability to handle fuzzy and grey data. It has been applied in areas such as product design and user experience optimization, industrial design (Zhang et al., 2024), mass customization (R. Wang et al., 2024), cultural and creative products (T. Wang et al., 2025), and remanufacturing engineering (Zhou and Ke, 2022). Thus, GRA serves as a fundamental tool for design optimization and efficiency improvement. It has been used to analyze the geometric parameters of a mouse and its appearance characteristics (Zhang et al., 2024), seasonal fluctuation factors in retail (Wang et al., 2020), or user ARs weighting for wickerwork lamps (Wang and Yang, 2023). Some studies further explored time-weighted GRA (Wang et al., 2020) and its integration with QFD (Wang and Yang, 2023), offering new perspectives for requirements analysis. Recently, QFD has attracted attention in both service design (Chen et al., 2024) and product design (Liang et al., 2024), with significant progress in integrating various decision-making analysis methods. Moreover, advances in digital technology and AI have accelerated functional iteration in product design. The shortened design cycle and dynamic user feedback make time an essential variable in data-driven design analysis. Previous research has focused on the temporal nature of online reviews and user-generated content. For example, incorporating time-aware models into sentiment analysis or opinion dynamics analysis (Asadabadi et al., 2023). However, these studies have primarily focused on the evolution of sentiment over time, rather than utilizing data over time to inform the ranking of specific feature weights, which has led to theoretical and methodological gaps. Therefore, this study formulates a time-aware requirement weighting mechanism, in which Grey Relational Analysis (GRA) is employed as the analytical tool to operationalize temporally evolving functional requirements.

Finally, after computing weights for ARs and FRs, it is necessary to map them jointly onto design features so that

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design decisions are guided primarily by users' affective expectations while remaining constrained by functional requirements. ARs capture users' emotional demands regarding product appearance and overall perceptual experience and thus constitute the principal driver of design decisions. By contrast, FRs reflect users' explicit concerns about concrete functional attributes and are used in this study primarily as auxiliary correction and constraint factors. Because ARs and FRs differ substantially in semantic expression and in how their weights are derived, modeling or mapping them separately tends to weaken the consistency of affect-driven design and may even yield results that deviate from users' true perceptual expectations. QFD provides a systematic and traceable framework for transforming the "voice of the customer" into "engineering characteristics." Through the House of Quality (HoQ), QFD clarifies and quantifies the correspondence between user requirements and DFs (Lee et al., 2020). Gai et al. (2024) proposed a QFD model for e-commerce platforms by combining social network analysis and group decision making, emphasizing the impact of expert opinion consistency on design priorities. Chen et al. (2024) propose a Kano-QFD approach driven by online reviews, which takes into account the impact of financial situation on determining the importance of users' ARs, providing a new perspective on demand processing. Wang and Yang (2023) integrated GRA with fuzzy QFD to design and develop improved willow lamps, deriving optimal parameters through emotional factor analysis and verifying the combined model's reliability. Although QFD has been widely applied as a mapping model in design research, most studies focus on the separate mapping of either FRs or ARs. Few studies have integrated both ARs and FRs into a unified QFD matrix for comprehensive analysis. QFD was originally developed to map user requirements (WHAT) to DFs (HOW), a logic that aligns closely with the decomposition process of ARs and FRs (Lyu et al., 2022). Therefore, this study adopts Quality Function Deployment (QFD) as a unified structured mapping mechanism within the WHAT–HOW framework. With ARs and FRs, QFD collectively guides the screening and prioritization of design features, thereby achieving a synergistic mapping that is emotionally driven and functionally constrained.

To summarize, this study incorporates both ARs and FRs into the user requirements side of the QFD matrix. This is done to achieve a systematic transformation of DFs. In comparison with the single mapping method, this framework can maintain the scientific nature of prediction and evaluation. Furthermore, it can directly serve the product design process, which is more grounded and operable.

2.3 AI image generation

Although user requirements have been effectively identified and mapped in previous studies, conceptual product design still relies mainly on designers. Even experienced designers face limitations of subjective judgment in practice (Wu et al., 2022). With advances in image generation technology, more researchers now employ it to simplify and accelerate the design process (Yuan et al., 2024). Yang et al. (2023) combined KE and Factor Analysis (FA) to process user comments, integrating them with Diffusion Modeling (DM) to design product forms meeting users' ARs. Image-generation technology has become an active research field. Text-to-image technologies have gained significant attention in product design research (Z. Wang et al., 2025). Models such as SDM, Generative Adversarial Networks (GAN), and Midjourney have demonstrated high performance in high-fidelity image generation and iterative denoising. They have been widely applied in product design rendering (Du et al., 2024), automotive front design (Yuan et al., 2024), and bionic design (Z. Wang et al., 2024a). When handling high-dimensional and complex data, the GAN model exhibits heightened sensitivity to natural-language input, demonstrating superior learning capacity that enables it to generate higher-resolution images and more intricate shapes with precision (Zang et al., 2024). However, GANs require substantial training time and computational resources, and their optimization process often lacks stability (Z. Wang et al., 2024b). Midjourney focuses on artistic quality and creativity, enabling rapid image generation and large-scale colour variation exploration (Wu et al., 2024). However, as a closed-source model, Midjourney cannot be customized or fine-tuned; users are restricted to its existing architecture and capabilities. Thus, for APD images requiring domain-specific or highly specialized styles, the model may fail to fully satisfy user requirements, thereby constraining its flexibility. SDM is a versatile text-to-image cross diffusion model that can generate realistic images from text inputs and is suitable for Text-to-Image and image-to-image generation tasks (Z. Wang et al., 2025). In product design, Kang and Zhao (2025) integrated the SDM with a Gated Recurrent Unit–Attention mechanism to realize automatic optimization of ceramic product design. Lee et al. (2024) proposed fine-tuning image generation models based on user preferences to visualize diverse interior design styles. SDM has also been preliminarily applied in architecture and product design, yielding positive results. Given its demonstrated potential in shape generation and structural optimization, this study employs the SDM in conjunction

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with user-requirement analysis to conduct an in-depth exploration and application of APD form optimization.

2.4 VR eye-tracking techniques

The primary criterion for evaluating design sketches lies not in their realism or quantity but in their alignment with consumers' psychological expectations. Therefore, eye-tracking experiments are widely used to ensure scientific rigor and minimize subjective bias (Li et al., 2024; Y. Wang et al., 2023). Such experiments have been applied in product aesthetics (Y. Wang et al., 2023), humanoid robot appearance design (Li et al., 2024), colour preference (Huang and Lin, 2020), visual aesthetic preference (Luo et al., 2022), multimodal integration (Kang et al., 2024), and related domains. However, traditional eye-tracking experiments were mostly limited to 2D spaces, failing to capture participants' observation behaviours in 3D environments (Bigne et al., 2024). To address this limitation, scholars introduced VR technology into eye-tracking experiments, enriching both application scenarios and research methodologies (Moreno-Arjonilla et al., 2024). Serrano et al. (2024) demonstrated that different VR navigation methods—such as natural walking and teleportation—significantly influence gaze distribution and observation duration. These papers indicate that VR eye-tracking experiments, by offering an immersive 3D interactive environment, overcome the limitations of 2D spaces. VR eye-tracking offers unique advantages for product design evaluation and user experience optimization. Moreover, 3D images are more realistic and can better display the product's actual appearance (Liu and Yang, 2022). Accordingly, this study employs VR eye-tracking as the validation method, immersing users in controlled environments to minimize external interference during design evaluation.

3. Methodology

To assist enterprises in designing product forms that meet user requirements under resource constraints, this paper proposes an integrated hybrid product design method that combines user review analysis with SDM. This method is suitable for products that emphasize both aesthetics and functionality as key selling points. Such products can accurately identify various user requirements, providing effective image references and theoretical guidance for product design. As illustrated in Fig. 1

Stage 1: data collection, user reviews, and image data for specific products are collected from the internet and preprocessed for subsequent stages. Stage 2: User requirements analysis: In the first part, Kansei words (KW) are extracted from the cleaned online reviews to identify Kansei factors that reflect key ARs. A method combining synonym clustering and RCA is employed to calculate the weight of each AR. The Kansei factors were then cluster analyzed using FA for subsequent use. In the second part, FRs are extracted from the review data, and an initial screening is performed based on word frequency. An innovative approach using the GRA under the influence of time weight is applied to calculate the weights of product features. Subsequently, QFD was utilized to translate ARs and FRs into product functional requirements, and their weights were calculated. Stage 3: image generation, the ARs weights and FRs weights obtained in the second stage are integrated to generate final prompts for use in SDM. Stage 4: image filtering, the generated images undergo preliminary screening by industrial design experts to create conceptual sketches. A combined subjective and objective method is applied for secondary screening, and the product design sketch with the highest score is selected as a reference for designers to finalize the product design. Stage 5: experimental verification, the improved design model is compared with the original model. Validation is conducted using a VR-based eye-tracking experiment to ensure the effectiveness and practicality of the design improvements.

3.1 User reviews collection

In stage 1, user reviews and images of branded products are collected from the internet and processed. To ensure the reliability of the data, it is necessary to select a suitable data source. In this study, the mainstream e-commerce platform "JD.com" is selected for this purpose. Reviews on this platform typically contain rich KW and user feedback, which provides data support for subsequent affective analysis. After data collection, data cleaning and processing are carried out. The review data may contain noise and invalid information, such as duplicate reviews, negative comments, promotional content, and irrelevant discussions. NLP techniques are employed to filter out such invalid information. Subsequently, the Jieba library is used for part-of-speech tagging and the extraction of valid user feedback. The processed

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review data serves as input for the next stage to support the extraction and analysis of ARs and FRs.

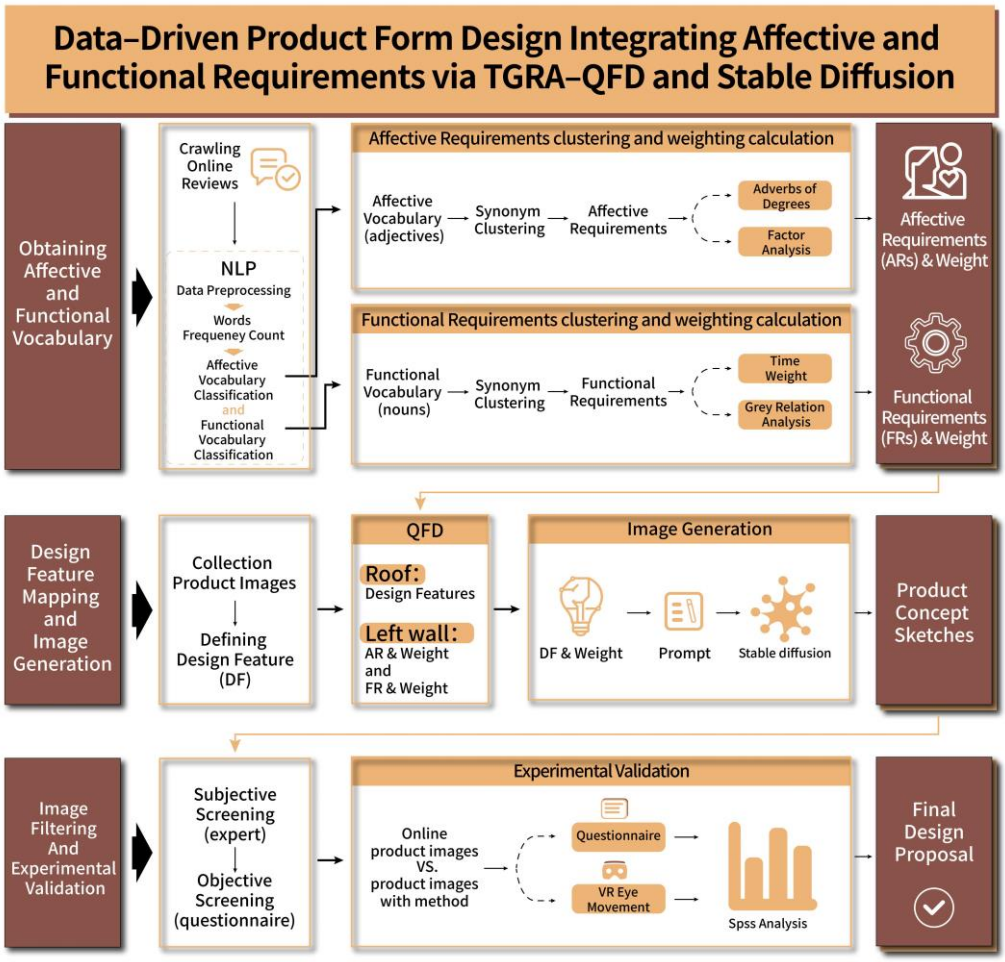


Fig. 1 Flowchart of product form design method based on user reviews and SDM. This figure shows the design process from data acquisition to VR eye movement verification, with key aspects including emotional and functional need weighting calculation and AI image generation.

3.2 Affective requirements

In stage 2, we extract users' ARs from online reviews and quantify their relative importance. First, review texts are harvested from e-commerce platforms via web-scraping. Because KW are primarily conveyed through adjectives, we compile a Kansei lexicon by identifying the most frequently occurring and semantically relevant positive adjectives in the corpus. The crawler extracts these high-frequency affective terms, which encapsulate users' perceptions of and aspirations for the product.

When selecting products, consumers often make subjective evaluations based on their experiences. In such cases, consumers use emotionally charged adjectives such as "delicate" and "beautiful" to express their feelings. Once extracted, these KW are grouped, labeled, and merged into a representative Kansei attribute (KA) (Lee et al., 2020). Let $A = A_1, A_2, \ldots, A_n$ represents the set of KA for the product, and $B = B_1, B_2, \ldots, B_n$ denotes the linguistic term set for measuring KA using a K-point scale.

Subsequently, the KW were clustered. The process of extracting data from online reviews follows the computational steps outlined in (Ren et al., 2023). The formulas and calculation steps are as follows:

Let $E = E_1, E_2, \ldots, E_n$ represent the set of reviews for all substitute products, and E_m denote the total collection of reviews. $E_m(A_n)$ refers to the number of reviews E_m that contain KW belonging to the KA A_n. $E_m(A_n^k)$ indicates the number of reviews $E_m(A_n)$ that are rated at the level B_k and include KW. Therefore, the formula is calculated as Eq. (1).

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$$E_m(A_n) = \sum_k E_m(A_n^k) \tag{1}$$

For KA A_n, the proportion of reviews containing KA A_n that are rated at level B_k is denoted as $E_m(A_n^k)$. Thus, as shown in Eq. (2).

$$A_n^k = \frac{E_m(A_n^k)}{E_m(A_n)} \tag{2}$$

In addition, this study uses degree adverbs (such as "very," "slightly," etc.) to precisely determine the weight of each Kansei factor. These adverbs help to more finely quantify the intensity of consumers' emotions. The study also employs the cumulative probability distribution for each level of the product's ARs to describe the degree of consistency between attribute levels and consumer preferences.

The possibility distribution encodes the assumed preference structure across intensity levels:

$$\mu(B_k) = \frac{K-k}{K-1} \tag{3}$$

The distribution is normalised to yield a probability mass function:

$$P_n(B_k) = \frac{\mu(B_k)}{\sum_{j=1}^{K} \mu\left(B_j\right)} \tag{4}$$

Cumulative probabilities aggregate the likelihood of achieving at least the specified level:

$$P_n^*(B_k) = \sum_{i=1}^{k} P_n\left(B_i\right), k = 1,2,\ldots,K \tag{5}$$

For each AR $A_n^* \in A^*$ of the product, based on the ARs of the product, the score for each AR A_n^* can be calculated, denoted as S_n. Where $P_n^*(B_k)$ represents the probability that the affective vocabulary level B_k satisfies the user's requirements. As shown in Eq. (6).

$$S_n = \sum_{k=1}^{K} A_n^k * P_n^*(B_k) \tag{6}$$

FA will be applied to cluster ARs and identify key factors that significantly influence users' affective experiences. In summary, through this calculation method, the Kansei factors that users are most concerned about during product use and their corresponding weights can be identified, providing valuable insights for subsequent design decisions.

3.3 Functional requirements

To ensure that the products generated by SDM exhibit similar overall appearance characteristics, it is necessary to input the functional features of the product—such as a single camera, grip design, etc.—during the image generation process. This allows for an effective enhancement of the image generation outcome.

Therefore, this section first utilizes the user review texts of products collected from shopping websites. Since functional vocabulary is primarily in the form of nouns, high-frequency, meaningful nouns are selected as functional feature terms to construct a dictionary. Next, the Time-weighted Grey Relational Analysis (TGRA) is employed to calculate the weight of each functional feature term. By introducing a time decay factor, this method addresses the issue of weight calculation associated with the temporal variation of user requirements.

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3.3.1 Time-weighted Grey Relational Analysis (TGRA)

During the initial process of extracting nouns, the timestamps of the reviews are recorded, and a weighting method is applied to reflect the changing trend of user demand over time. To capture the temporal decay of user demand, we introduce a time-decay factor into the GRA weighting scheme. Assume that the review dataset covers the period from the earliest observation year Y_{min} to the latest observation year Y_{max}. For an individual review issued in year y ($Y_{min} \leq y \leq Y_{max}$), the year distance (YD) is calculated as follows (Asadabadi et al., 2023):

$$YD = Y_{max} - y \tag{7}$$

The final time weight formula is shown in Eq. (8).

$$W_Y = e^{-\frac{YD^2}{Y_{max}-Y_{min}}} \tag{8}$$

In the subsequent GRA, these year weights $W_Y(j)$ are used only as column weights in Eq. (16) and are not pre-multiplied to the raw yearly counts. Within-year scaling and reference series. For each year j, let

$$\tilde{x}_i(j) = \frac{x_i(j)}{max_i x_i(j)} \in [0,1], x_r(j) = 1 \tag{9}$$

This makes each year comparable by taking the within-year maximum as the reference (ideal) value. The Grey theory provides a metric called the 'gray relational coefficient' to compare the closeness between design elements and the reference vocabulary (reference series) to calculate the weight values of design elements, with the following calculation steps (T. Wang et al., 2025) To establish comparison levels: Suppose X denotes an $m \times n$ matrix to collect assessments of the performance levels of n-dimensional elements in m alternatives, where $x_i(k)$ denotes the element k of alternative i, m is the number of a series of alternatives, and n denotes the number of criterion attributes. As shown in Eq. (10).

$$X = \begin{bmatrix} x_1(1) & \cdots & x_1(n) \\ \vdots & \ddots & \vdots \\ x_m(1) & \cdots & x_m(n) \end{bmatrix} \tag{10}$$

To set the maximum rating scale corresponding to the elements in the reference series, which is also similar to the ideal solution. As shown in Eq. (11).

$$x_r = \{(maxx_i(j_i) \mid j \in \bar{J}), (minx_i(j_i) \mid j \in \bar{L})\} \quad i = 1,2,\dots,m \tag{11}$$

where $\bar{J}$ denotes the set of benefit terms, and $\bar{L}$ denotes the set of cost terms. The deviation matrix is used to calculate the 'degree of grayness':

$$\Delta = \begin{bmatrix} \Delta_{1r}(1) & \cdots & \Delta_{1r}(n) \\ \vdots & \ddots & \vdots \\ \Delta_{mr}(1) & \cdots & \Delta_{mr}(n) \end{bmatrix} \tag{12}$$

where $\Delta_{ir}(j) = \left|\left|x_i(j) - x_r(j)\right|\right|$, $x_i(j)\ and\ x_r(j)$ represent the jth element of the target and reference series, respectively. Then, the attribute-based grey correlation coefficient is calculated:

$$\gamma\left(x_i(j), x_r(j)\right) = \frac{\Delta_{min} + \xi\Delta_{max}}{\Delta_{ir}(j) + \xi\Delta_{max}} \tag{13}$$

$$\Delta_{min} = \min_{\forall i} \mathrm{m}_{\forall j} \parallel x_i(j) - x_r(j) \parallel \tag{14}$$

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$$\Delta_{max} = \max_{\forall i} \mathrm{m}_{\forall j} \parallel x_i(j) - x_r(j) \parallel \tag{15}$$

where a distinguishing coefficient $\xi \in (0,1)$ is set at the 0.5 value. The grey correlation degree is calculated:

$$\Gamma\left(x_i(j), x_r(j)\right) = \sum_{j=1}^{n} W_{Y(j)}\, \gamma\left(x_i(j), x_r(j)\right) \tag{16}$$

3.4 Establish the mapping model with QFD

AR reflects perceptual expectations, while FR represents explicit functional concerns. Treating them separately would lead to fragmented design decisions. Integrating both into a unified QFD matrix enables the simultaneous consideration of perceptual and functional drivers within a single WHAT–HOW mapping structure. Therefore, this study uses QFD as a mapping model to transform ARs and FRs into cue words that can be understood by SDMs, thus improving the accuracy of the generated images. The absolute weight $A_{ij}$is obtained by multiplying the customer requirements weight by the relevant matrix's correlation symbol. This weight is then used to calculate the weight of the QFD engineering features, and the absolute weight of the engineering features is computed and weighted. The calculation of absolute weights requires formalization to derive the relative weights R_{ij}, as shown in Eq. (17). and Eq. (18):

$$A_{ij} = \sum_{i=1}^{n} W_i * R_{ij} \tag{17}$$

$$R_{ij} = \frac{A_{ij}}{\sum_{j=1}^{m} A_{ij}} \tag{18}$$

W_i represents the importance value of the i-th user requirement; R_{ij} represents the relationship value between the i-th user requirement and the j-th engineering feature.$i = 1,2, \ldots, j = 1,2, \ldots m$ represents the index for DFs.

3.5 Product image generation

In stage 3, by combining the ARs weights and FRs weights determined, a Prompt is generated to be provided to SDM. This method, by integrating prompt engineering techniques, ensures that the generated design prompt reflects both users' ARs and FRs, thus more comprehensively guiding product development. Specifically, the weights of each feature are integrated and averaged through a weighted mean to generate a comprehensive prompt. This prompt will be used by designers to maintain a balance between ARs and FRs in the product design. It will assist in generating conceptual design sketches through generative models like SDM. This step not only generates design proposals that integrate both user affective experience and functionality but also lays the foundation for subsequent expert reviews and design optimization.

3.6 Image filtering

In stage 4, to ensure the scientific objectivity of the screening process, this study will conduct further selection by distributing questionnaires to drone users, designers, and undergraduate and graduate students. The clustered Kansei factors will then be used to design questionnaires for different groups, collecting additional subjective feedback from users. Combining this feedback with expert opinions, the design sketches will undergo secondary screening. The sketches with the highest weighted scores will serve as a design reference for the final product design stage.

3.7 Product evaluation using a VR eye-tracking device

In stage 5, this paper verifies the effectiveness of the improved design model through a virtual reality eye-tracking

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experiment and compares it with the original model. Before the experiment, we'll create a virtual reality environment for participants to view the pre- and post-improvement APD design models. Eye-tracking technology will collect data on users' attention distribution and visual focus during the experiment. After the experiment, this data will help us determine which design is more appealing to users. Ultimately, the experimental results will offer a scientific basis for the design's final verification and guide future design optimization.

4. Case study

The APD's versatility really appealed to customers and made its sales volume go way up (Q. Wang et al., 2023). In this paper, we use the design of APDs as an example to validate the effectiveness of the proposed method. Currently, the functionality and visual characteristics of most drones on the market have a positive impact on consumer purchasing intent. Particularly in online shopping, users primarily base their purchasing decisions on the product's appearance and available features. Therefore, functional characteristics and design elements have become key factors in consumers' decisions to purchase drones.

4.1 Extraction of affective requirements

This study uses user reviews from JD.com as the source of ARs. To ensure data diversity, APDs were selected, and Python-based web scraping collected five years of reviews (2019–2024), treating each year as a distinct stage. This multi-year approach compensates for the absence of clear product-generation boundaries. After removing duplicate texts, a total of 10,912 review texts were collected. The Jieba library is then used to tag and filter adjectives in the review sentences. This process yielded 561 affective adjectives. A mixed quantitative-qualitative screening was applied to refine KW. Adjectives with a frequency of over 50 were counted, while we removed words that merely indicate good or bad without actual meaning, as well as pejorative terms. The remaining positive adjectives are used to construct a Kansei lexicon containing 32 KW (Table 1). Synonyms were clustered to identify key ARs, merging the lexicon into eight representative KWs.

To assess the product quality, adverbs preceding the Kansei words are extracted along with the adjectives from the review data. In this context, based on existing research (Ren et al., 2023), all adverbs were classified into six levels. Let $B = B_1, B_2, \ldots, B_6$represent the scale used to measure affective intensity. In this set, $B1$ (e.g., "especially," "particularly") represents the highest level, B5 (e.g., "slightly," "a bit") represents the lowest level, and B6 represents other or none.

The evaluation can be accumulated for each AR at each level across all products. Table 1 lists the number of comments at each level for the eight ARs of the drone products. Based on the method described for extracting affective data from online reviews, Eq. (1) can be used to systematically establish the preliminary KA related to the drone reviews and ARs $A = A_1, A_2, \ldots, A_8$.

The probability that a given ARs level matches consumer preferences is calculated using Eq. (3)-(5), with the results detailed in Table 2. Subsequently, the AR weights of the APDs can be calculated using Eq. (6), and the final weights w_n are normalized by S_n. The final results of the AR weights of the aerial photography APDs are shown in Table 2. The extracted ARs served as Kansei evaluation factors. Samples were drawn from the four top-selling drone products on JD.com, and front, left, top, and side views were assessed using a five-point Likert scale questionnaire.

Table 1 The number of reviews at various levels in the KA

Kansei word	B_1	B_2	B_3	B_4	B_5	B_6	Sum	Weight
Swift	643	481	36	39	7	1724	2930	0.219
Stable	423	424	41	43	0	1631	2562	0.162
Simple	591	767	88	90	6	2761	4303	0.256
Lightweight	169	117	23	13	0	519	841	0.057
User-friendly	77	84	10	11	0	339	521	0.031
Exquisite	439	410	33	31	0	1405	2318	0.162
Compact	216	334	26	14	4	1118	1712	0.100
Cool	32	29	4	0	0	123	188	0.012

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Table 2 The probability that each level aligns with the consumer's preferences

B	B1	B2	B3	B4	B5	B6
$\mu(B_k)$	1	0.8	0.6	0.4	0.2	0
$P_n(B_k)$	0.3333	0.2667	0.2	0.1333	0.0667	0
$P_n^*(B_k)$	1	0.6667	0.4	0.2	0.0667	0

A total of 59 questionnaires were received. After screening and removing invalid responses, 38 valid questionnaires were retained for FA. Cronbach's alpha = 0.884, indicating high reliability. Additionally, the Kaiser-Meyer-Olkin (KMO) test of sampling adequacy and Bartlett's test of sphericity were conducted. The KMO test result was 0.908, indicating that FA is appropriate, and Bartlett's test had a significance level of 0.000, showing that the data follow a spherical distribution. This means the variables are not independent, satisfying the conditions for FA. SPSS software was employed to perform factor analysis on the dataset. Principal component analysis was adopted for clustering the 8 Kansei image words, followed by orthogonal rotation of the factor structure using the varimax rotation method. The rotated component matrix is presented in Table 3 (only entries with an absolute factor loading value $\geq$ 0.5 are reported).

In order to obtain the relative weights of the extracted Kansei factors, a unit-weighted approach was adopted following (DiStefano et al., 2009). In this approach, each Kansei word belonging to the same factor is treated as a formative indicator, and its contribution to the factor is assumed to be equal. Accordingly, the average weight of all Kansei words within each factor was computed to represent the raw factor weight. Based on AR's weight, the computed scores for the drone are shown in Table 3.

Table 3 Rotated component matrix table

Kansei words	1	2	3	Kansei factor	Weight(ω)
Lightweight	.823				
Compact	.754			Portability (AR1)	0.346
Swift	.565				
Stable		.804			
Simple		.659		Suitability (AR2)	0.413
User-friendly		.622			
Cool			.849	Appearance (AR3)	0.240
Exquisite			.544		

4.2 Extraction of functional requirements

After the reviews were collected, noun parts of speech in the drone reviews were tagged and filtered, while removing pronouns and words with no actual meaning. Among the remaining nouns, five meaningful terms are selected, and their word frequencies are calculated after merging synonyms and transforming them into functional features. During the filtering process, each term is also tagged with a temporal variable.

4.2.1 Time-weighted Grey Relational Analysis (TGRA)

In this stage, GRA is used to obtain the weight of each term, helping designers identify key FRs. At the same time, a temporal variable is introduced. The final result is calculated using Eq. (7) and Eq. (8), as shown below: 2024:1.0, 2023:0.819, 2022:0.449, 2021:0.165, 2020:0.041, 2019:0.007. The number of functional terms containing time variables is shown in Table 4. However, there are fewer comments from earlier years. This is because sales websites often remove some outdated comments and only keep a part of the valid ones, leading to this issue. The above calculation results were used as the weights for the Grey Relational Analysis (GRA). Subsequently, the grey relational degrees of the five functional requirements were computed, enabling quantitative assessment of user concern for each functional feature extracted from the reviews. To examine the effect of incorporating temporal information, the conventional GRA was also applied to the same dataset for comparison. The resulting weights obtained from both methods are reported in Table 5. As shown in the table, although the overall ranking order of the functional requirements remains unchanged—indicating

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the robustness of the weighting results—the weight distribution produced by TGRA exhibits a noticeably higher variance than that of the conventional GRA. By incorporating temporal weights, TGRA differentiates persistently emphasized functional concerns from those with diminishing relevance over time, thereby enhancing the discriminative power of functional requirement weighting. In contrast, the equal-weight treatment in traditional GRA leads to a more uniform distribution with lower dispersion, which may weaken the sensitivity of subsequent design feature mapping.

Table 4 Functional terms with the introduced temporal variable

Time Weights(W_t)	**Fuselage stability**	**Long-lasting**	**Strong flight capability**	**Great manoeuvrability**	**Camera function**
1.000	1831	1332	103	2420	768
0.819	1430	934	39	1820	508
0.449	193	217	10	145	48
0.165	54	73	0	35	6
0.041	0	14	0	1	1
0.007	2	0	0	0	0

Six-year weights (2024 →2019:1.000,0.819,0.449,0.165,0.041,0.007) are used only as column weights in the aggregation stage of GRA; raw counts $C_{i,j}$ are not pre-weighted.

Table 5 Time-weighted GRA results for functional requirements

Weight	**F1**		**F2**		**F3**		**F4**		**F5**	
	γ	**Value**	**γ**	**Value**	**γ**	**Value**	**γ**	**Value**	**γ**	**Value**
0.403	γ11	0.333	γ21	0.526	γ31	0.412	γ41	0.663	γ51	1.000
0.330	γ12	0.333	γ22	0.501	γ32	0.404	γ42	0.695	γ52	1.000
0.181	γ13	0.333	γ23	1.000	γ33	0.380	γ43	0.812	γ53	0.590
0.067	γ14	0.333	γ24	1.000	γ34	0.353	γ44	0.624	γ54	0.490
0.016	γ15	0.333	γ25	1.000	γ35	0.350	γ45	0.333	γ55	0.350
0.003	γ16	0.333	γ26	0.333	γ36	0.333	γ46	1.000	γ56	0.333
TGRA grade	FR1	0.333	FR2	0.642	FR3	0.399	FR4	0.693	FR5	0.879
GRA grade	FR1	0.750	FR2	0.847	FR3	0.781	FR4	0.894	FR5	0.983
TGRA Relative weight	FR1	0.113	FR2	0.217	FR3	0.135	FR4	0.235	FR5	0.298

FR1=Strong flight capability; FR2=Long-lasting; FR3=Camera function; FR4=Fuselage stability; FR5=Great manoeuvrability.

4.3 QFD establishes a mapping association

Based on the experimental results (Table 5), the most representative APD-related terms were identified. Using the grey relational degree calculation results, the nouns were placed on the left side of the QFD as the focal points of user evaluation. DFs were then extracted from the images of the top-selling drone products and placed in the upper section of the QFD as functional attributes. Firstly, a focus group of 22 experts was invited, with the majority aged between 20 and 29 years, accounting for 40.91%. The group was evenly split by gender, with 11 males and 11 females, each representing 50% of the group. Educational backgrounds were primarily undergraduate and master's degrees, with 72.73% holding a master's degree and 27.27% holding an undergraduate degree. Prior to the relationship evaluation, a focus group consisting of 12 design experts was established. All experts specialized in intelligent product appearance design and had 5–15 years of professional experience, with eight holding a master's degree or higher. The group first defined a standardized relationship scale (Weak = 0.2, Moderate = 0.4, Strong = 0.6, Very strong = 0.8, and Extremely strong = 1.0). Rather than directly assigning numerical values, the experts assessed the relative influence levels of affective and functional requirements on each design feature based on their semantic understanding. These qualitative judgments were then systematically quantified according to the predefined scale to construct the QFD relationship matrix. Subsequently, the normalized weights of ARs and FRs were mapped to design features using Eqs. (17) and (18). The normalized results are presented in Table 6, and the resulting QFD matrix is shown in Table 7.

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Table 6 ARs and FRs Weighting Normalization Table

User Requirements	FR1	FR2	FR3	FR4	FR5	AR1	AR2	AR3
Weight	0.113	0.217	0.135	0.235	0.298	0.346	0.413	0.240
Normalized weight	0.057	0.109	0.068	0.118	0.149	0.173	0.206	0.120

4.4 Prompt definition

To ensure semantic consistency across generated images, all prompts were constructed in accordance with the syntax and weighting conventions of Stable Diffusion. Based on these conventions, a standardized prompt template for aerial drone image generation was formulated, as shown in Table 8. Stable Diffusion was selected as the backbone model for APD image synthesis because it allows the relative semantic emphasis of design-related descriptors to be modulated through prompt weighting, without assuming deterministic or directly interpretable control over specific visual attributes (Mo et al., 2024). Accordingly, prompt weighting in this study is employed as a relative semantic guidance mechanism. Its validity is assessed not through precise visual feature control, but through the consistency and comparability of visual tendencies across image sets generated under different affective-weight configurations, as examined in the subsequent evaluation sections.

4.5 Image filtering

In this section, the focus group was first invited to evaluate the 200 APD images that had been generated subjectively, and selected 17 of them. To improve transparency and reduce arbitrariness in image filtering, a structured expert-based qualitative evaluation procedure was adopted. These criteria served as common reference dimensions to constrain individual judgment. Experts employed comparative judgment rather than numerical scoring to exclude images exhibiting obvious geometric distortion, implausible structural configurations, or clear perceptual inconsistency. In addition, ARs and FRs were also used as shared reference dimensions to further constrain individual judgment. Each expert retained approximately 30–40 representative and design-feasible images. The retained subsets were then compared, and images selected by multiple experts were considered to exhibit higher qualitative consensus. Based on this inter-expert overlap, 17 images were selected as the final set for subsequent analysis. These 17 images were then objectively evaluated in conjunction with the perceptual factors obtained from the FA and the questionnaire. Furthermore, to ensure that prompts derived from this methodology produced aerial-drone images aligned with users' affective needs, the random seeds of the 17 selected images were fixed, and three variants per seed were generated under three prompt conditions: (a) DFs plus weights, (b) DFs only, and (c) no DFs. This yielded 17 groups of three corresponding drone designs. To control questionnaire length, the 10 most representative groups were chosen. A questionnaire referencing the three clustered affective needs was administered using a seven-point Likert scale; 44 valid responses were collected. The questionnaire's Cronbach's alpha was 0.989. Mean scores were: design features + weights = 47.8785; design features only = 45.454; prompts only = 43.793.

Table 7 QFD matrix

	Weight	Design Features					
		DF1	DF2	DF3	DF4	DF5	DF6
FR1	0.057	1.0	0.6	0.8	1.0	0.2	
FR2	0.109		1.0	0.6	0.6		0.8
FR3	0.068			0.6			1.0
FR4	0.118	0.6	1.0			0.8	
FR5	0.149	0.2	0.4	1.0	1.0	0.8	
AR1	0.173	1.0	1.0		0.2	1.0	0.2
AR2	0.207	0.6	0.8	1.0	0.4	0.8	1.0
AR3	0.120	0.6	1.0		1.0	0.6	0.8
Absolute Important		0.53	0.78	0.51	0.51	0.64	0.49

DF1=Foldable arms; DF2=Lightweight and compact design; DF3=Four sensors; DF4=Quadcopter; DF5=Integrated fixed leg landing gear; DF6=single-lens camera.

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Table 8 Prompt definition

Prompt Definition	**3D, C4D render, Unreal Engine, Octane render, an aerial drone, (DF 1: Weight), (DF 2: Weight), ..., white background.**
Final Prompt	3D, C4D render, unreal engine, octane render, an aerial drone, (Foldable arms: 0.53), (Lightweight and compact design:0.78), (Four sensors:0.51), (Quadcopter:0.51), (Integrated fixed leg landing gear:0.64), (single-lens camera:0.49), white background,
Negative prompt	NSFW, (worst quality:2), (low quality:2), (normal quality:2), lowres, normal quality, ((monochrome)), ((grayscale)), (ugly:1.331), (duplicate:1.331), (morbid:1.21), lowers,extra digit,

Using the obtained common factors and their weights, combined with the 17 images selected in the preliminary screening, a 7-point Likert scale was used to design a questionnaire. The target respondents were experts in industrial design and experienced drone users. The questionnaire was distributed again, where participants were asked first to observe the overall design of the 17 drones. Afterwards, they were instructed to write down their initial impressions of the drones (including first feelings, desire to buy, etc.), followed by filling out the questionnaire. A total of 6 valid responses were received (Due to the lengthy duration of this section (30-60 minutes) and the presence of some unusable questionnaires, the number of valid questionnaires is relatively low.). The results of the questionnaire were calculated and analyzed. The final scores indicated that drone models 5 and 15 received the highest ratings. Three professional designers were invited to replicate and refine the creative images.

4.6 Experimental

In everyday shopping scenarios, users typically encounter two options: online shopping and physical shopping. However, many drone retailers now use 3D models displayed on their websites to attract users. In addition, VR devices, compared to traditional desktop-based eye-tracking equipment, cause less interference in the subjects' vision during experiments, providing a more objective reflection of their preferences. Therefore, this study ultimately uses VR equipment for the eye-tracking experiment, with aerial drones as the experimental stimuli, and records the eye movement data to derive the conclusions. Two design sketches were generated via image generation tools, based on ARs and FRs extracted from user comments, and the two best-selling drone designs were chosen as stimulus materials. They were converted into 3D models for subsequent VR eye-tracking experiments.

4.6.1 Stimulus

To ensure experimental validity, the two best-selling models among the four drones initially selected were retained and modelled for comparison with the modified designs generated by the methodology proposed herein (Fig. 2).

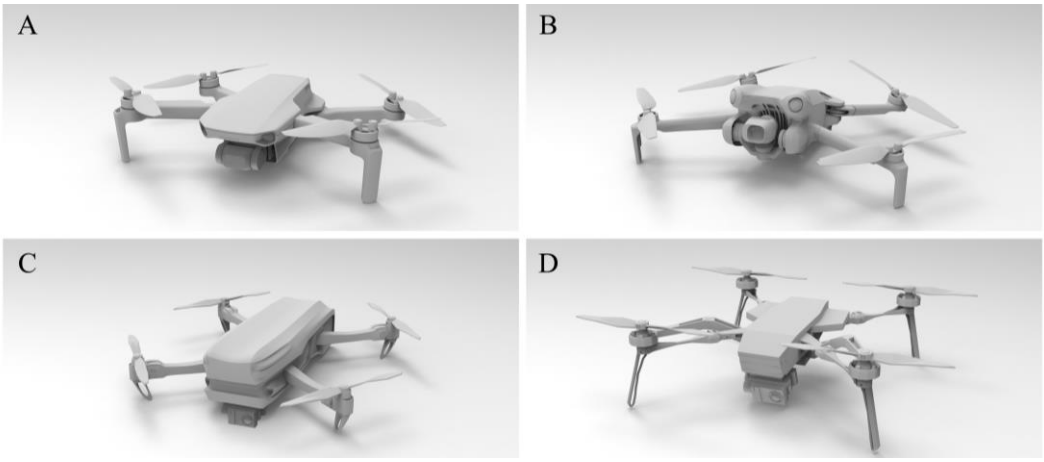


Fig. 2 Partial APD form design results of image generation. A: The first existing APD model. B: The second existing APD model. C: The first APD model generated and modelled via SDM. D: The second APD model generated and modelled via SDM

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4.6.2 Participants

Participants in this study were primarily drawn from the 18–34 age group. A total of 24 participants were recruited for the experiment (12 males and 12 females, mean age 22.2 years, SD = 9.8). including both undergraduates and postgraduates, were recruited as participants. All participants had normal or corrected-to-normal vision, signed the informed consent form, and did not engage in strenuous exercise or consume caffeine on the day of the experiment or prior. They all expressed their willingness to cooperate in completing the experiment. The eye-tracking experiment utilized the Ergo Lab series VR eye tracker. The setup included a main unit and a display screen. The stimulus was presented in the center of the VR headset screen, with a white background. The Ergo Lab 3.0 software installed on the computer assisted the eye tracker in recording the experiment, as well as collecting, displaying, and exporting the eye movement data for each participant. The laboratory temperature was maintained at 26°C, with a humidity of 40%.

4.6.3 VR Eye-tracking

To analyze the eye-tracking data, Areas of Interest were defined. The drones generated by SDM were labeled as generated design (GD), while those retrieved from JD were labeled as existing design (ED). Each stimulus was assigned its corresponding AOI and identification number. Using Ergo Lab, the data were exported and processed in SPSS to calculate several AOI metrics for two stimulus groups: fixation duration (FD) and fixation count (FC).

The experiment was conducted for four stimuli, with participants using VR headsets. Each stimulus was presented individually in a 360-degree panoramic image with a white background. To eliminate the influence of irrelevant variables on the study, the product color was excluded, and participants were introduced to the experimental procedure and relevant precautions before the experiment.

A total of 24 participants were scheduled in a pre-determined, gender-balanced order and tested individually. Upon arrival, each participant first read the on-screen instructions and then previewed the four drone renderings to counterbalance order effects. After a one-minute resting period with eyes closed, participants entered a three-minute evaluation phase. Each trial began with a five-second exposure to a uniform grey 360° background image, followed by the sequential presentation of the four drone models in individually randomized order. Every model was displayed for ten seconds and then replaced by a five-second grey blank screen; this 10-s stimulus/5-s inter-stimulus cycle was repeated three times per model, yielding twelve total exposures. Participants were instructed to “freely explore the design with their gaze while keeping the head still”; gaze was tracked continuously throughout the session. The procedure is illustrated in Fig. 3

At the end of the experiment, we invited the subjects to fill out a subjective questionnaire using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) to evaluate four APD prototypes in terms of portability, usability, and aesthetics, followed by a “preferred design” question to identify overall user preference.

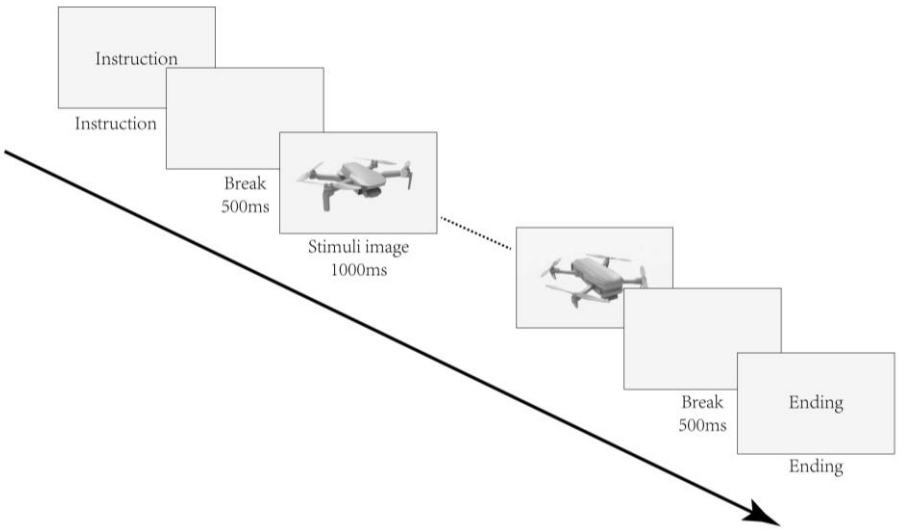


Fig. 3 Experimental Procedure of the VR Eye-Tracking Experiment

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4.6.4 Result

Visual attention differences among four drone stimuli (comprising two existing designs (ED): E1, E2; and two generated designs (GD): G1, G2) and between two drone parts (wing vs. fuselage) were analyzed using SPSS. A 2 × 4 within-subjects repeated-measures analysis of variance (ANOVA) was employed, with a sample size of N = 24 and four repetitions per cell. Additionally, a linear mixed-effects model was used to estimate participant-wise random intercepts and slopes; the pattern of means and significance levels derived from this model was consistent with those of the ANOVA. Greenhouse‑Geisser corrections were applied whenever the assumption of sphericity was violated. The results are summarised in Tables 9, Table 10, and Table 11.

Table 9 Analysis of experimental findings

Type	Model	Fixation Duration (FD)				Fixation Counts (FC)			
		Wing		Fuselage		Wing		Fuselage	
		M	SD	M	SD	M	SD	M	SD
ED	E1	6.58	2.74	10.10	3.99	23.17	8.99	33.33	14.03
	E2	2.64	1.95	13.48	4.93	9.33	5.80	43.21	14.08
GD	G1	4.57	3.35	12.48	3.52	15.00	8.48	42.38	10.93
	G2	6.49	2.61	10.67	3.70	61.42	13.77	38.04	10.19

The improvement stage (Existing vs. Improved) showed a significant main effect on total fixation duration, $F(1, 23) = 57.54$, $p < .001$, $\eta^2 = .714$, and on total fixation count, $F(1, 23) = 36.82$, $p < .001$, $\eta^2 = .615$. Simple-effects analyses with Bonferroni correction were conducted on the significant Model × Part interaction. In the wing region, GD (G1, G2 pooled) attracted longer total fixation duration than ED (E1, E2 pooled), mean difference = 0.92 s, 95 % CI [0.12, 1.72], $p = .021$, $\eta^2 = .25$, and 58 % more fixations, mean difference = 21.96, 95 % CI [15.8, 28.1], $p < .001$, $\eta^2 = .44$. At the fuselage region, no significant differences emerged (duration: mean difference = –0.21 s, 95 % CI [–1.42, 0.99], $p > .99$; count: mean difference = 1.94, 95 % CI [–5.2, 9.1], $p > .99$), indicating that the additional wing attention did not reduce fuselage attention.

Table 10 Comprehensive summary of eye-tracking results $N = 24$

Variable & Effect	df	F/t	p	η^2	Sig.
Fixation Duration					
Model	3,69	1.23	.305	.051	No
Part	1,23	57.54	<.001	.714	Yes
Model × Part	3,69	20.06	<.001	.466	Yes
Fixation Count					
Model	3,69	66.69	<.001	.744	Yes
Part	1,23	36.82	<.001	.615	Yes
Model × Part	3,69	98.28	<.001	.810	Yes

Sig. = significant at $p < .05$; η2= partial eta-squared. Greenhouse–Geisser corrections applied when $\varepsilon < .75$.

Table 11 Simple effects on wing and fuselage regions with Bonferroni-corrected pairwise comparisons.

Region	Existing	Improved	Mean Diff	p	95 % CI
Wing					
Duration (s)	4.61 (2.81)	5.53 (2.89)	0.92	.021	[0.12,1.72]
Fixation Count	16.25 (8.76)	38.21 (11.42)	21.96	<.001	[15.80,28.12]
Fuselage					
Duration (s)	11.79 (4.50)	11.58 (3.61)	–0.21	>.99	[–1.42,0.99]
Fixation Count	38.27 (14.06)	40.21 (10.57)	1.94	>.99	[–5.20,9.08]

As shown in Table 12, G2 achieved the highest ratings across all three dimensions, with mean scores of 4.96 for

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portability, 5.63 for usability, and 5.25 for aesthetics, yielding an overall average of 5.28. E1 ranked second (5.14), whereas E2 received the lowest overall evaluation (4.36). Among all dimensions, usability obtained the highest average score, indicating that flight stability and ease of control are the most critical factors influencing user perception.

Table 12 Mean scores of four UAV designs across evaluation dimensions $N = 24$

APD Design	Portability	Usability	Aesthetics	Overall Mean
E1	4.92	5.50	5.00	5.14
E2	4.38	4.50	4.21	4.36
G1	4.42	5.21	4.67	4.76
G2	4.96	5.63	5.25	5.28

Ratings were made on a 7-point Likert scale (1 = strongly disagree,7 = strongly agree). All values are the mean of the 24 valid questionnaires. All values are averages of 24 valid questionnaires.

To test the statistical significance of the differences among the four designs, a Friedman repeated-measures test was conducted. The results revealed X^2= 14.61, p = 0.002 (p < 0.01), confirming that the overall evaluations of the APDs differed significantly. Participants were able to clearly distinguish between design alternatives in terms of perceived quality and user experience.

In summary, the subjective evaluation results consistently indicate that G2 provides a superior overall user experience compared with the other design schemes. From an affective perspective, G2 exhibits higher visual appeal and perceptual coherence. From a functional–perceptual perspective, it more effectively conveys cues related to controllability and flight stability through its form characteristics. These findings suggest that perceived usability—rather than objectively measured performance—plays a dominant role in shaping users' evaluations of APD appearance during the early design stage. Accordingly, future product development should prioritize enhancing form-level cues associated with control feedback and stability, while preserving aesthetic balance and structural compactness, so as to strengthen users' overall perceptual acceptance and affective engagement.

5. Discussion

5.1 The contribution of the research

This study integrates image generation technology with user review analysis, focusing on APDs as the research subject. First, web crawlers collected online drone reviews from JD.com. Second, RCA was applied to objectively rank users' ARs. Third, temporal weights were introduced, and the TGRA was employed to rank FRs. Next, a QFD matrix was used to map ARs and FRs into DFs. Finally, SDM-based image generation and mixed subjective–objective evaluation verified that the resulting product designs effectively satisfied consumers' aesthetic preferences.

In this study, users' ARs and FRs are extracted and clustered separately from online reviews, which are mapped into DFs by combining KE techniques and transformed into product forms preferred by users. Previous KE studies employed AHP, entropy weighting, or TOPSIS to weight perceptual requirements and map them to DFs, bridging communication gaps between users and designers and facilitating the creation of refined product concepts. However, both perceptual weighting and sketch generation remain partly subjective, often leading to designs that fail to fully satisfy users' ARs. To address this issue, researchers have explored integrating image generation techniques into product design. Wang et al. (2025) proposed a multimodal generative design method constrained by emotional and functional factors, extracting affective–functional vocabularies to generate and optimize 3D product models via deep learning for more intelligent and personalized design. Yang et al. (2023) introduced image generation technology into product design and combined it with KE technology to help businesses obtain user-preferred product forms. Du et al. (2024) integrated image generation with KE to identify user ARs and generate corresponding product appearances, optimizing design processes and enhancing sustainable competitiveness. While these studies mitigate subjectivity in DF translation, they focus primarily on ARs and overlook FRs, which may result in designs lacking perceived functional plausibility and form-level feasibility from a user perspective. Moreover, the rapid pace of functional iteration in modern product design necessitates incorporating time as a variable (Asadabadi et al., 2023). Accordingly, this study extracts FRs from online reviews, determines their weights through TGRA, maps both ARs and FRs into DFs using QFD, and employs SDM to generate final concept sketches. A comparative summary with prior studies is shown in Table 13. Using KE-based methods, users' ARs and FRs

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were identified from online reviews to leverage implicit design knowledge and minimize expert subjectivity. They were incorporated into a QFD matrix, mapped to DFs, and used with SDM to generate numerous concept sketches, which were then screened through combined subjective and objective evaluations. Questionnaire surveys and VR eye-tracking experiments verified that product forms designed via the framework better satisfied users' aesthetic preferences than traditional designs. In summary, this study integrates image generation and KE methods, introducing requirement decomposition and time weighting to propose a practical product form design framework that helps enterprises create user-preferred products.

Table 13 Comparison of this study with previous articles

References	Data source	ARs	FRs	Design process
This Paper	Online review	RCA	TGRA	QFD+SDM
(Z. Wang et al., 2025)	Online review	KE+FA	Kano	SDM
(Du et al., 2024)	Online review	KE+FA+PLSR		SDM
(Yang et al., 2023)	Online review	KE + FA		DM
(T. Wang et al., 2025)	EGM	FA+GRA		Intuitionistic fuzzy VIKOR
(Chiu and Lin, 2018)	EGM	KE+FA+GRA		CDA

* EGM: Evaluation Grid Method, DM: Diffusion Model, CDA: Conceptual Design Automation

5.2 Limitations and future work

Although this study has obtained two form design schemes for APDs, there are still some limitations that need to be addressed.

Firstly, the methodology proposed in this study primarily focuses on product form design. However, other key factors, such as product colour and finish, can also have a significant impact on a user's purchasing decision. Owing to space constraints, these elements have not been comprehensively considered. Future research will investigate other crucial factors beyond ARs and FRs. Secondly, while this study utilized VR eye-tracking to minimize experimental interference and achieve enhanced data accuracy, the experiment still relied on static 360-degree images for product form evaluation. Subsequent investigations could integrate immersive technologies like Unreal Engine to examine product models in three-dimensional spaces. Future work will focus on extending this framework to color, texture, and material design dimensions, as well as validating the TGRA‑QFD‑SDM pipeline across multiple product categories. Thirdly, this study centers on affective-oriented appearance design and focuses on users' emotional perceptions and function-related associations elicited by APD product form during the early design stage. Accordingly, the functional requirements considered in this paper do not represent direct measurements of physical performance indicators such as flight stability or controllability; rather, they reflect function-related perceptual cues inferred by users from product morphology. These cues serve to provide reasonable functional constraints and design-consistency support for affective-driven design. Given the scope and methodological positioning of this study, experimental validation of the actual functional performance of APD devices was not conducted at this stage. Future work will incorporate performance-level validation methods, such as wind-tunnel testing or flight-control experiments, to achieve a multi-level integration from affective perception–oriented design to objective functional performance evaluation.

6. Conclusion

This study bridges the gap between affective‑functional requirement analysis and AI-driven product design frameworks. While previous studies have mainly focused on either emotional or functional dimensions in isolation, this study systematically integrates both aspects into a unified, data-driven process. By integrating TGRA, QFD, and the SDM, the framework enhances interpretability, traceability, and creativity in product form design. This study proposes a product design method that integrates user review analysis with SDM. The method analyzes user reviews to explore the relationship between ARs, functional requirements, and user preferences, and then generates images through SDM. Conceptual sketches that meet user requirements are selected from the generated images. This framework provides designers and enterprises with innovative guidance for product form development. It improves design efficiency, enhances the potential competitiveness and sustainability of product concepts, and broadens opportunities for integrating

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resources, technologies, and research methods. The contributions of the proposed method to Product form design are as follows:

(1) By applying SDM to APD product design, this study rapidly generates a large volume of high-quality images, replacing designers' hand-drawn sketches and thereby diminishing the influence of subjectivity.

(2) This study decomposes user requirements into affective and functional categories, thereby enabling a more precise articulation of specific demands for APD products.

(3) This study proposes a TGRA to mitigate the bias arising from the temporal decay of user comments.

(4) In the experimental stage, the study employs VR-based eye-tracking to assess the validity and scientific rigor of the proposed method. Compared with conventional screen-based eye-tracking experiments, the VR setting minimizes screen-related interference, thereby yielding more objective and precise data.

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